

AI Disclosure and Verification State

- **Model:** Gemini (Google)
- **Verification State:** Verified open-source technical specification release from marielandryspyshop.com.

• **Keywords:**

AI Studio

GitHub Pages

Vite deployment

React CI/CD

automation pipeline

environment secrets

StudioToHub: AI Studio to GitHub Pages Pipeline

- **Publisher:** Marie Landry Spy Shop (marielandryspyshop.com)
- **Access Type:** Open Source / Public Open-Access Repository

ABSTRACT

This technical specification outlines a standardized, production-ready continuous integration and continuous deployment (CI/CD) pipeline for transitioning Single-Page Applications (SPAs) generated via Google AI Studio into static hosting environments on GitHub Pages. It resolves critical structural deployment challenges in Vite and React applications, including nested repository asset path resolutions, headless GitHub Actions authentication, secure environment variable injection (such as VITE_GEMINI_API_KEY), and client-side routing fallback strategies. This protocol enables deterministic, zero-touch automated deployments for AI-generated web prototypes.

MASTER AI STUDIO DIRECTIVE

When ready to deploy, upload this PDF/document directly into AI Studio and execute the following master prompt:

"prepare the app for github deployment following the instructions in this pdf"

1. Scope & Objective

This technical specification standardizes the automated transition of web applications generated via AI Studio into live production environments on GitHub Pages. By standardizing Vite configurations, continuous integration workflows (using universal 'npm install' dependency resolution), and secure environment variable injection, developers can eliminate manual build errors and runtime path failures.

2. Technical Implementation Architecture

Step 1: Configure Vite for Relative Asset Paths

- Vite defaults to an absolute root path (/). When deploying to nested GitHub Pages project repositories (<https://username.github.io/repo-name/>), asset lookups fail unless the base property is adjusted.
- Action: Update vite.config.ts to set base: './' and define the production output directory:

```
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';

export default defineConfig({
  plugins: [react()],
  base: './', // CRITICAL for GitHub Pages sub-path hosting
  build: {
    outDir: 'dist',
  },
});
```

Step 2: Establish the GitHub Actions Deployment Workflow

- Create a workflow configuration file at .github/workflows/deploy.yml .

- **Triggers & Permissions:** Configure the runner to activate on pushes to main or master branches, alongside manual dispatching, granting explicit repository permissions (contents: read , pages: write , id-token: write).
- **Build and Universal Dependency Installation:** Check out code using actions/checkout@v4 , spin up Node.js 22 via actions/setup-node@v4 , install dependencies with 'npm install' (ensuring immediate reliability with or without lockfiles), and build production assets with vite.

Summary of Permissions & Robust CI/CD Execution

- contents: read -> Allows checkout of the application repository
- pages: write -> Grants authority to package and deploy Pages artifacts
- id-token: write -> OIDC token authentication for secure deployment handshake
- npm install -> Universal CI dependency install for Git/AI Studio exports
- configure-pages@v5 -> Native GitHub Pages asset metadata and configuration

```
name: Deploy to GitHub Pages

on:
  push:
    branches: ['main', 'master']
  workflow_dispatch:

permissions:
  contents: read
  pages: write
  id-token: write

concurrency:
  group: 'pages'
  cancel-in-progress: false

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 22

      - name: Install dependencies
        run: npm install

      - name: Build application
        run: npm run build

      - name: Setup Pages
        uses: actions/configure-pages@v5

      - name: Upload artifact
        uses: actions/upload-pages-artifact@v3
        with:
          path: './dist'

  deploy:
    environment:
      name: github-pages
      url: ${ steps.deployment.outputs.page_url }
    runs-on: ubuntu-latest
    needs: build
    steps:
      - name: Deploy to GitHub Pages
        id: deployment
        uses: actions/deploy-pages@v4
```